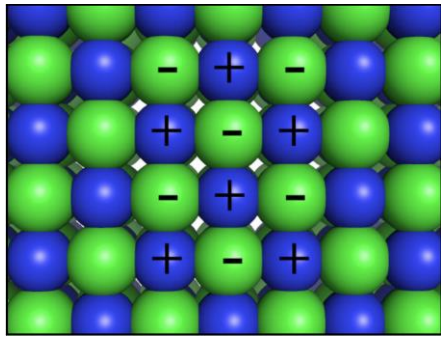
Chapter 3

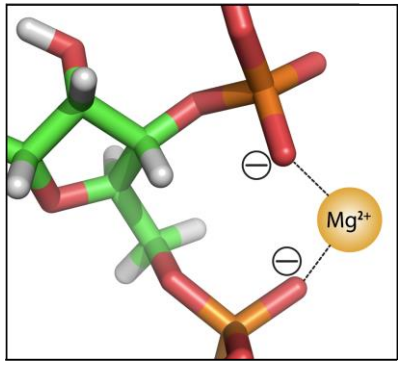
Molecular Interaction

Interactions between molecules

Ion-ion interaction: $V \propto 1/r$
Ion-dipole interaction

(r is the distance between the molecules)

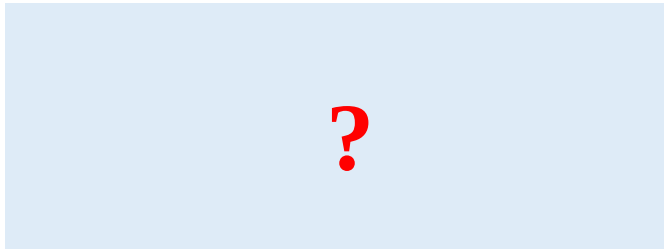




A **van der Waals interaction** : $V \sim 1/r^6$

Hydrogen bonds are especially strong dipole–dipole interactions between molecules that have hydrogen bonded to a highly electronegative atom, such as O, N, or F.

Polar molecule: a molecule with a permanent electric dipole moment.

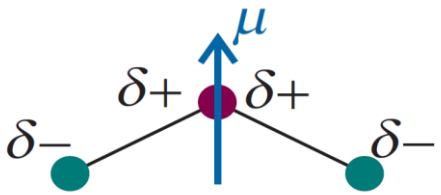


Ozone, O_3

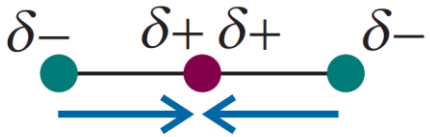


Carbon dioxide, CO_2

Polar molecule: a molecule with a permanent electric dipole moment.

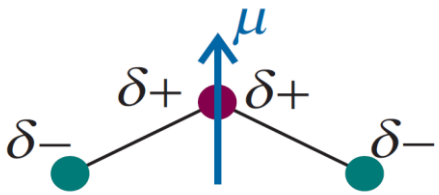


Ozone, O_3

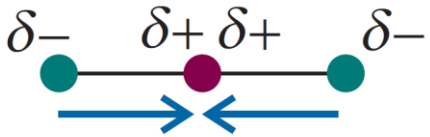


Carbon dioxide, CO_2

Polar molecule: a molecule with a permanent electric dipole moment.



Ozone, O_3



Carbon dioxide, CO_2

An applied electric field can distort a molecule as well as align its permanent electric dipole moment to generate an

Induced dipole moment: $\mu^{\star} = \alpha \mathcal{E}$

polarizability ($C^2\,m^2\,J^{-1}$)

Strength of applied electric field

Polarizability volume: $\alpha' = \dfrac{\alpha}{4\pi\varepsilon_0}$

has dimensions of volume;
↔ HOMO-LUMO gap: gap small, α′large

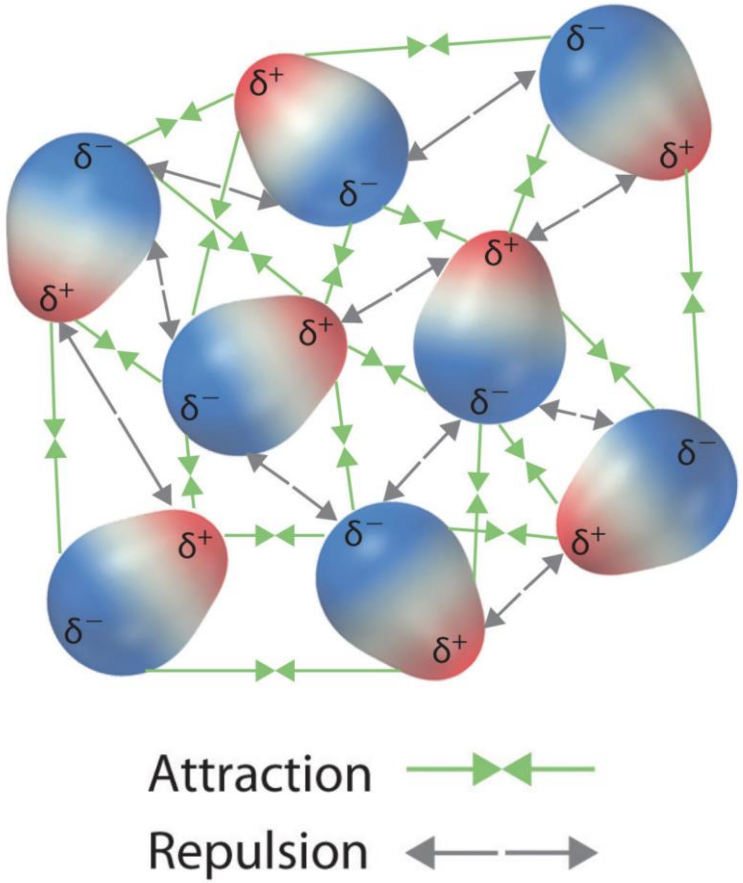
ε_0: Vacuum permittivity

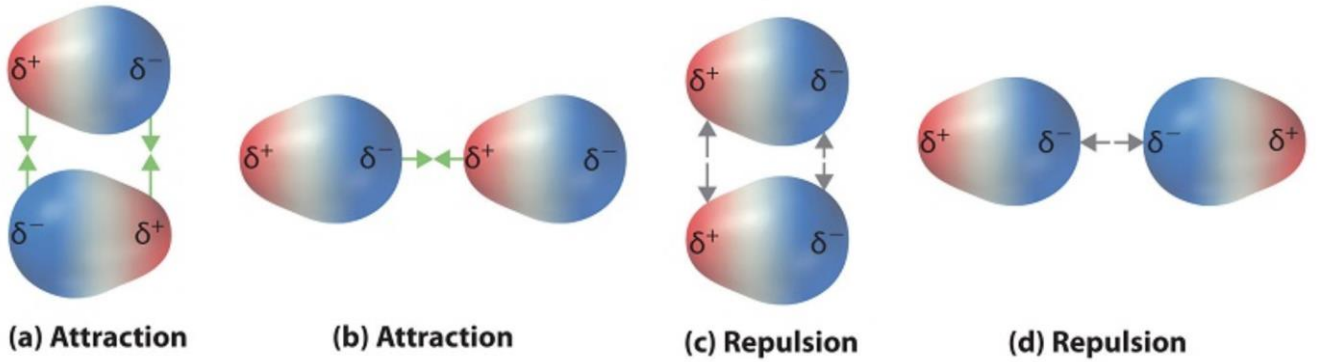
Interactions between molecules

A **van der Waals interaction** is the attractive interaction between closed-shell molecules: ~ $1/r^6$ (r is the distance between the molecules)

- **Electrostatic force**: Dipole-dipole interactions
- **Induced force**: Dipole-induced-dipole interactions
- **Dispersion force**: Induced-dipole-induced-dipole interactions (**London interaction**)

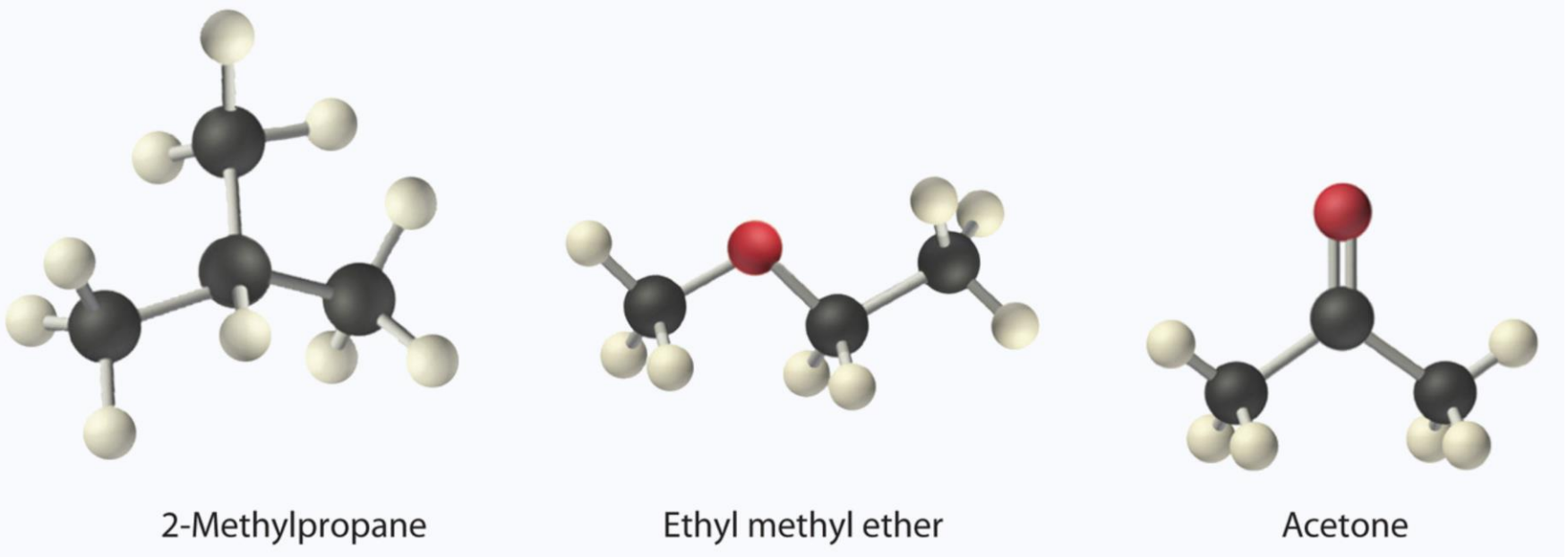
Dipole–dipole interactions





Favorable **Unfavorable**

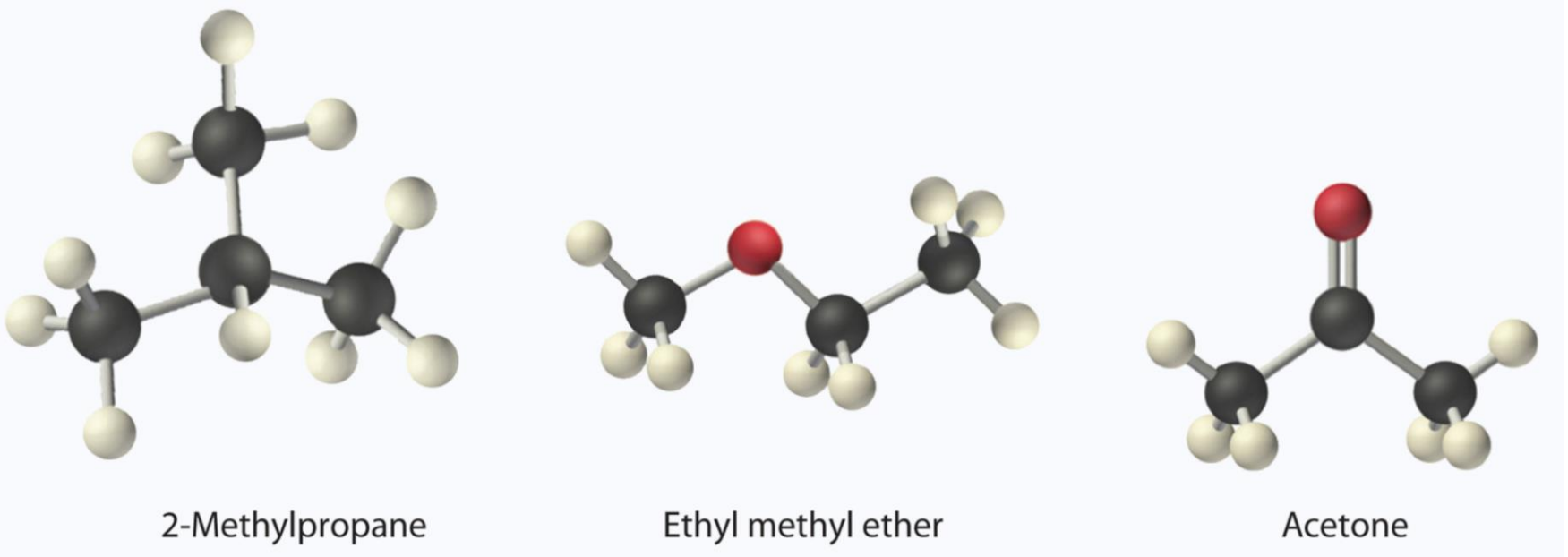
Dipole–dipole interactions



Boiling point

Dipole moment

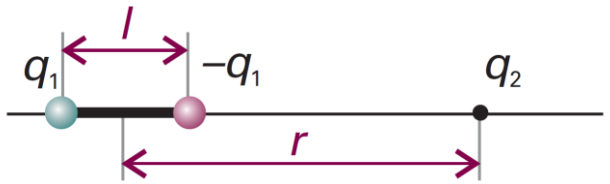
Dipole–dipole interactions



Boiling point	-11.7 C	7.4 °C	56.1 °C
Dipole moment	0.13 D	1.17 D	2.88 D

Interactions between dipoles

A point dipole ↔ A point charge

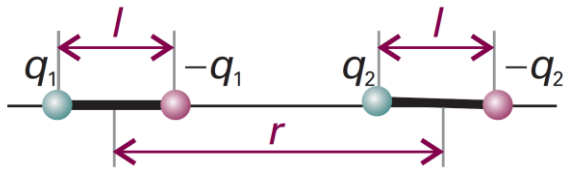


$\mu_1 = q_1 l$

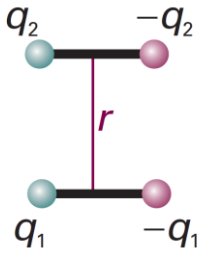
$l \ll r$

$$V = -\frac{\mu_1 q_2}{4\pi\varepsilon_0 r^2}$$

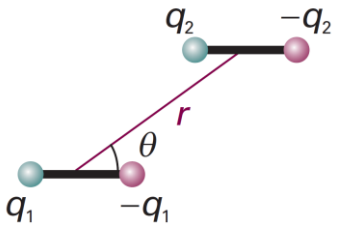
Dipole ↔ Dipole



$$V = -\frac{\mu_1 \mu_2}{2\pi\varepsilon_0 r^3}$$



$$V = \frac{\mu_1 \mu_2}{4\pi\varepsilon_0 r^3}$$



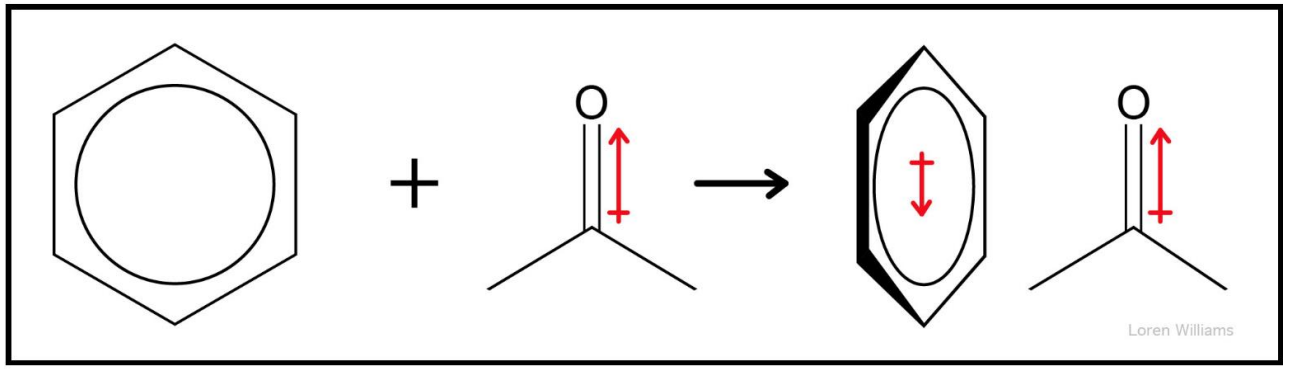
$$V = \frac{\mu_1 \mu_2 f(\theta)}{4\pi\varepsilon_0 r^3} \qquad f(\theta) = 1 - 3\cos^2\theta$$

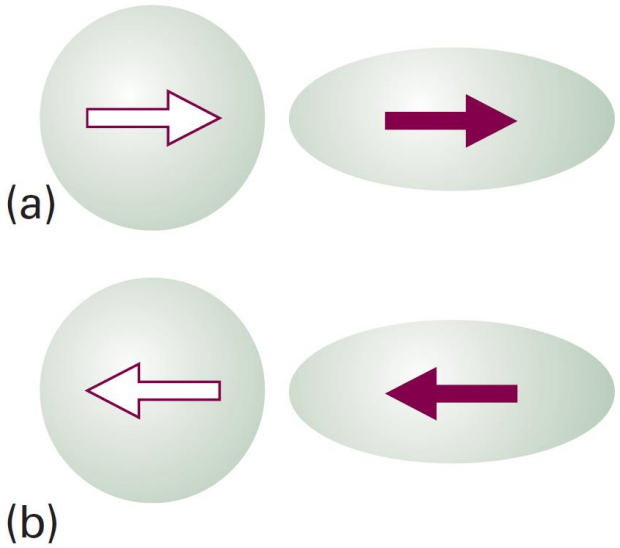
The average potential energy of two rotating molecules:

$$\langle V \rangle = -\frac{C}{r^6} \qquad C = \frac{2\mu_1^2 \mu_2^2}{3(4\pi\varepsilon_0)^2 kT}$$

Keesom interaction, the first of the contributions to the van der Waals interaction

Dipole–induced-dipole interactions



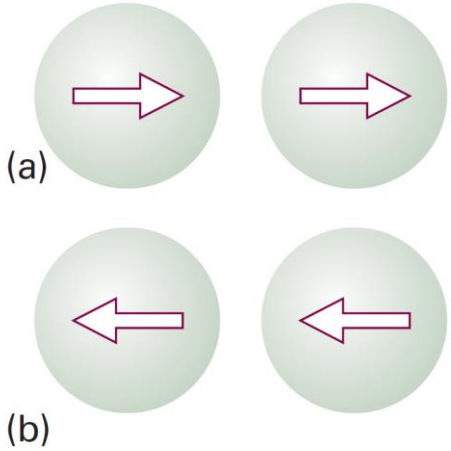


Molecule 2 Polar molecule 1 with μ_1

The polarizability volume of molecule 2

$$V = -\frac{C}{r^6} \qquad C = \frac{\mu_1^2 \alpha'_2}{4\pi\varepsilon_0}$$

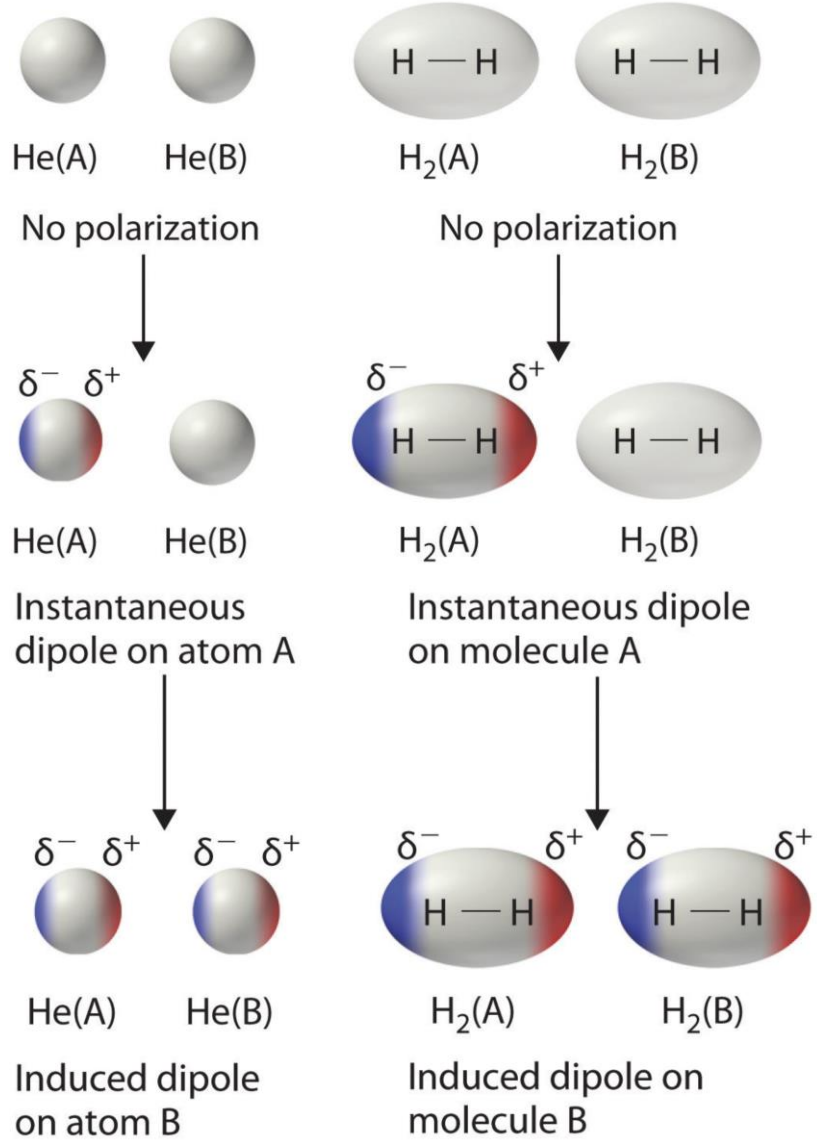
Induced-dipole–induced-dipole interactions



Ionization energy

$$V = -\frac{C}{r^6} \qquad C = \frac{3}{2}\alpha'_1\alpha'_2\frac{I_1 I_2}{I_1 + I_2}$$

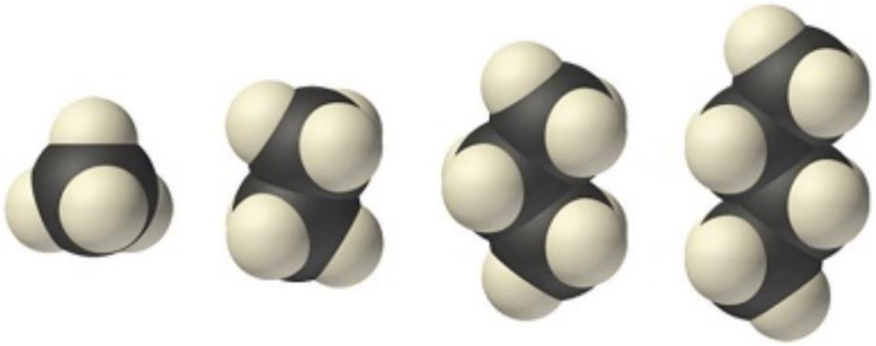
Dispersion interaction (London force), the third contribution to the van der Waals interaction



Induced-dipole–induced-dipole interactions

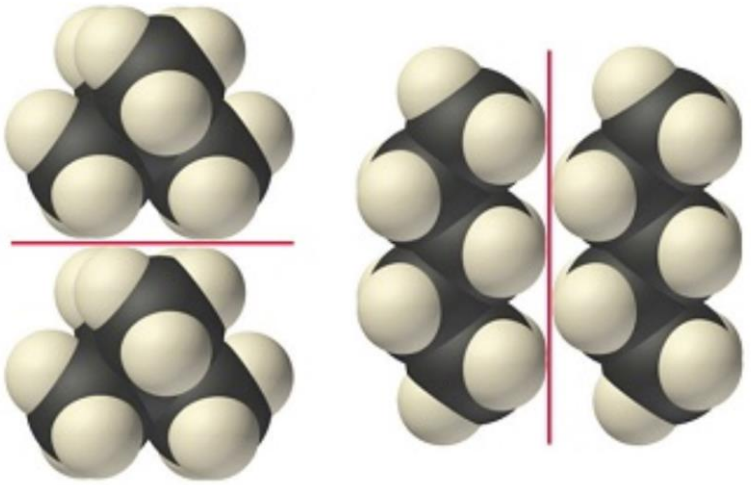
$$V = -\frac{C}{r^6} \qquad C = \frac{3}{2}\alpha'_1\alpha'_2\frac{I_1 I_2}{I_1 + I_2}$$

- Mass ↑, surface area ↑ → London force ↑;
- Depend on molecular shape



Methane 16 g/mol −161.5°C | Ethane 30 g/mol −88.6°C | Propane 44 g/mol −42.1°C | *n*-Butane 58 g/mol −0.5°C

(a) Increasing mass and boiling point



2,2-Dimethylpropane (neopentane) 72 g/mol, 9.5°C | *n*-Pentane 72 g/mol, 36.1°C

(b) Increasing surface area and boiling point

The total attractive interaction:

Depends on the identity of the molecules

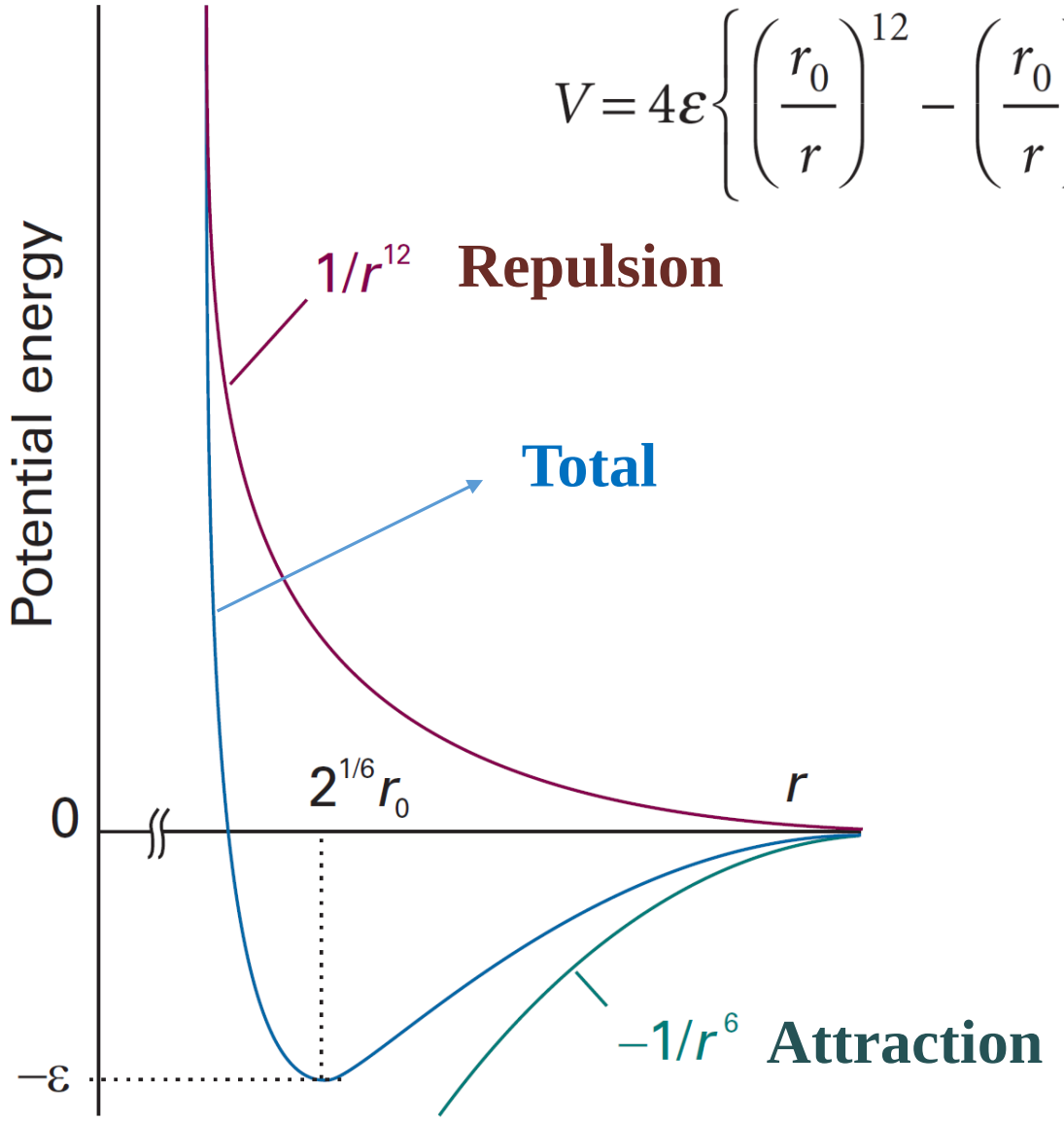
$$V = -\frac{C_6}{r^6}$$

Repulsive interaction:

Nuclear and electronic repulsions + rising electronic kinetic energy

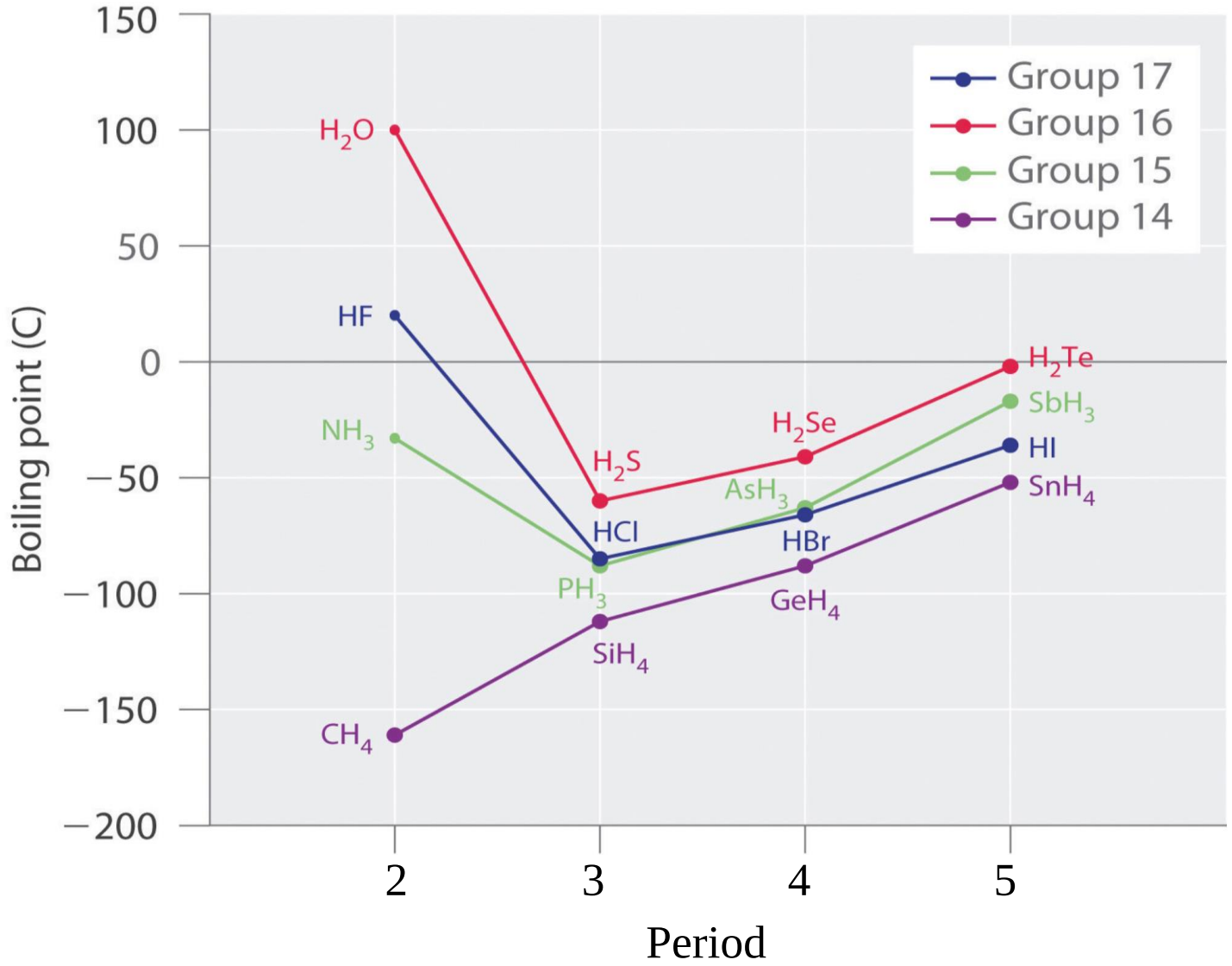
Total interaction ⟶

$$V = 4\varepsilon\left\{\left(\frac{r_0}{r}\right)^{12} - \left(\frac{r_0}{r}\right)^{6}\right\}$$



Lennard-Jones potential

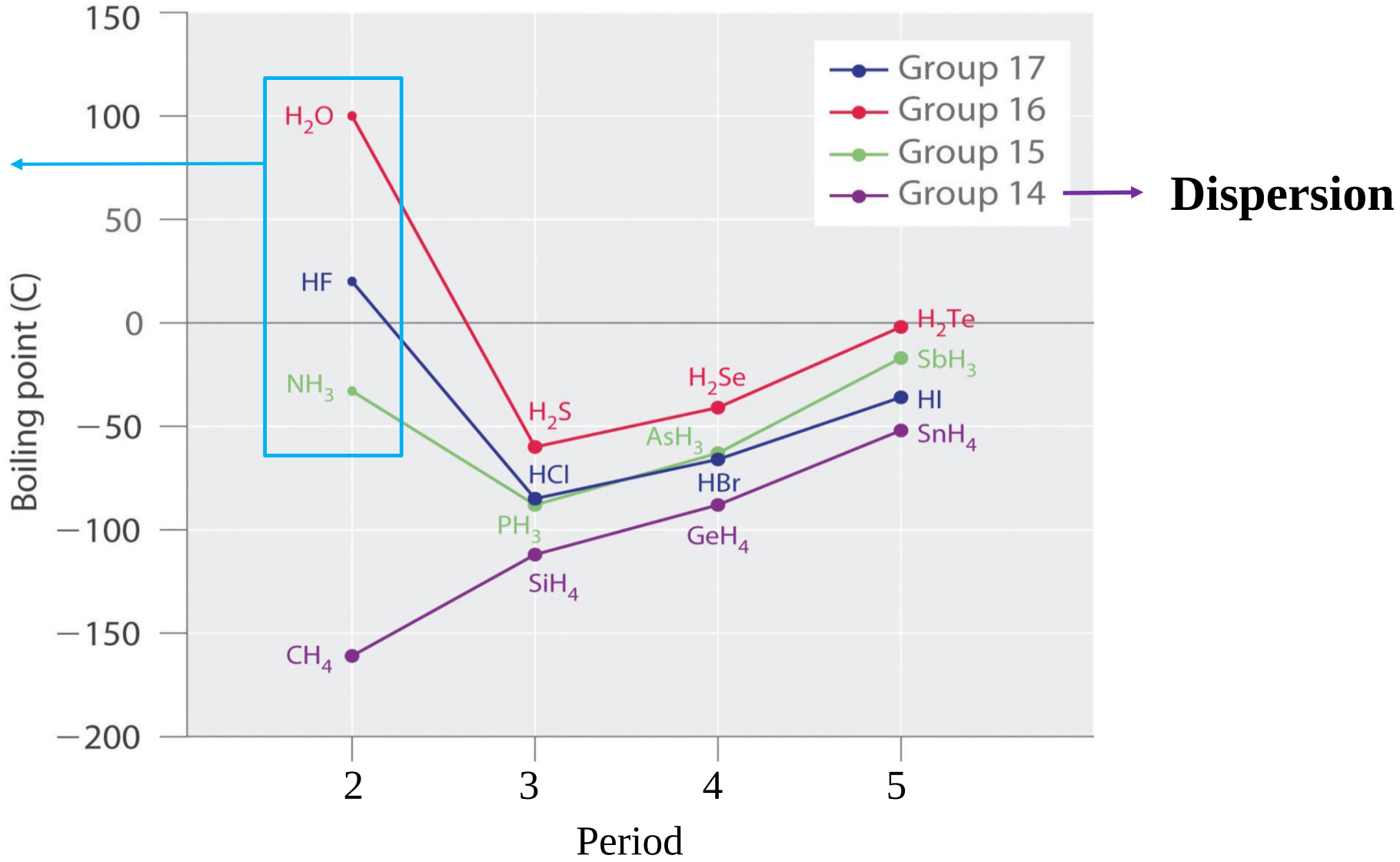
Interactions between molecules



Molecular Interaction

Interactions between molecules

Hydrogen bond



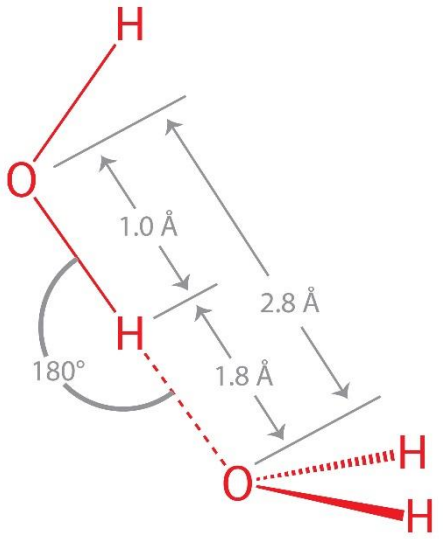
Dispersion

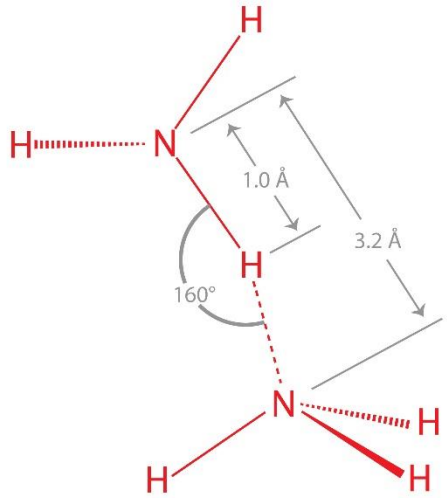
Hydrogen bonding A–H…B ~ 20 kJ/mol

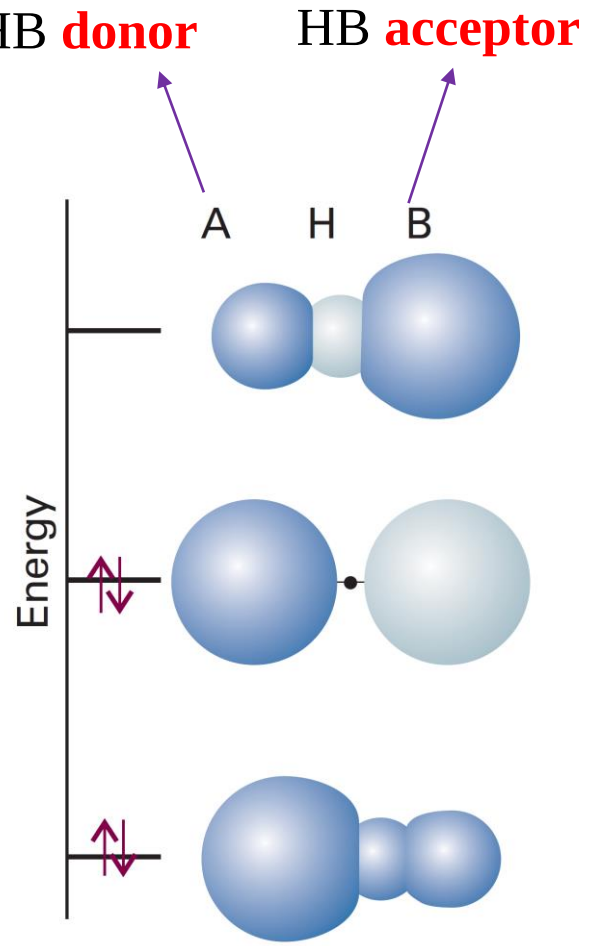
A and B: highly electronegative elements, N, O, F, or Cl^-…
B: Possesses a lone pair of electrons.

The combination of large bond dipoles and short dipole–dipole distances ⇒ very strong dipole–dipole interactions

Linear or nearlinear arrangement

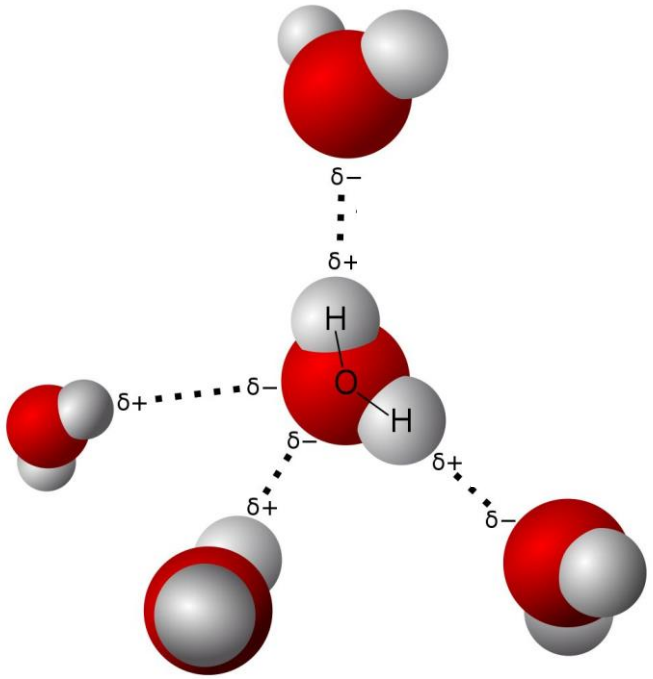


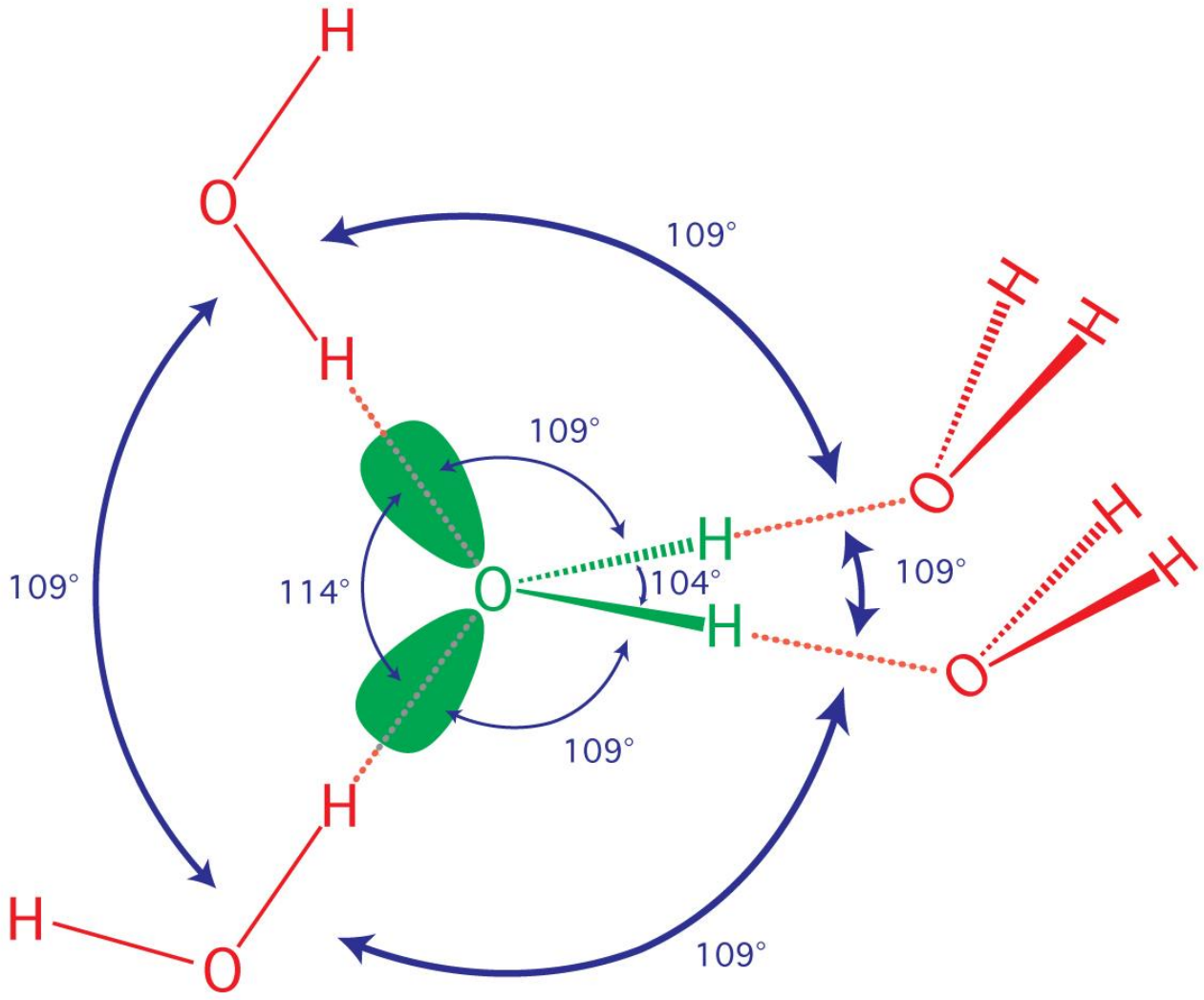




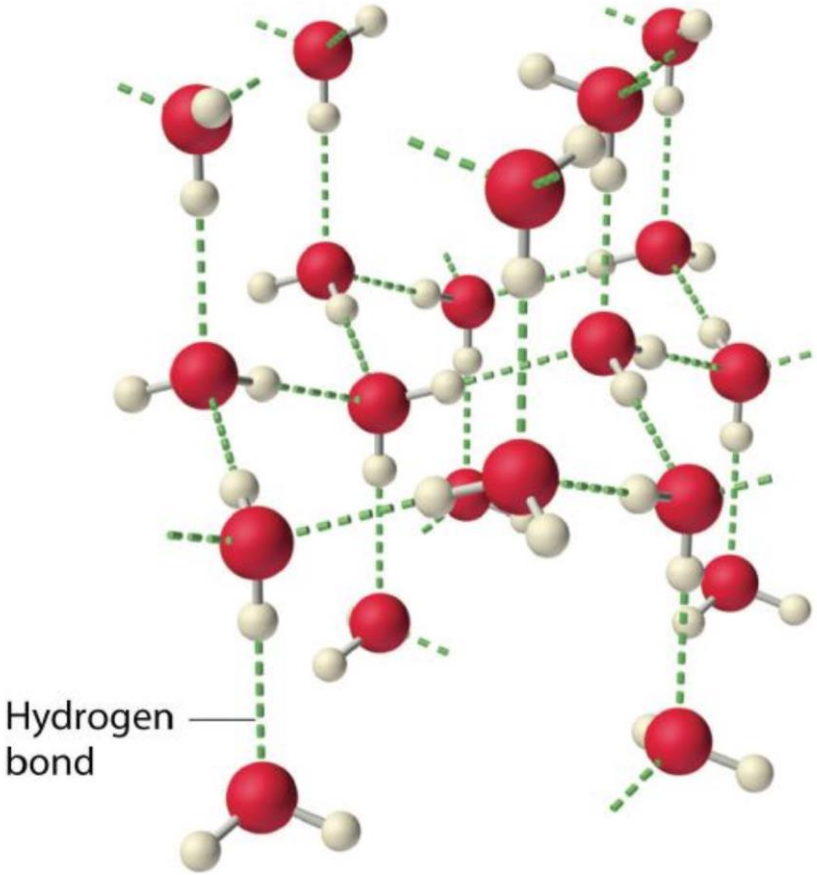
Symmetric hydrogen bond: F–H…F^-; Unsymmetric hydrogen bond: A–H…B

Hydrogen bonding





Hydrogen bonding

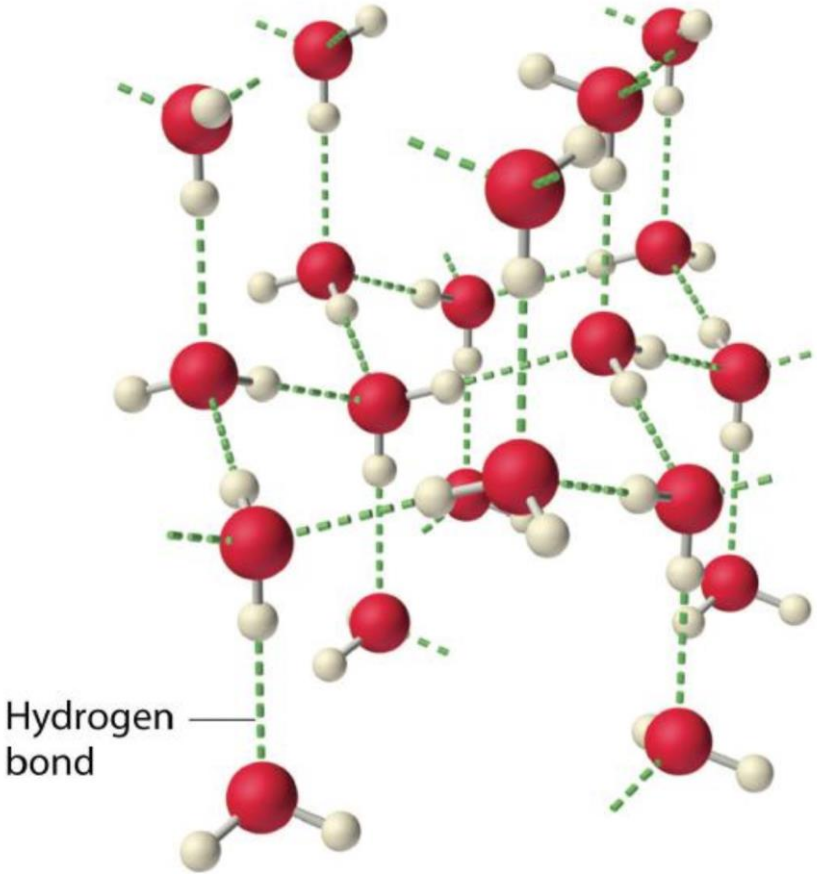


The Hydrogen-Bonded Structure of Ice

(Open, cagelike structure)

Less dense structure of ice leads that ice floats on water rather than sinks.

Hydrogen bonding



The Hydrogen-Bonded Structure of Ice

(Open, cagelike structure)

In **liquid water**, the hydrogen bonds are continually broken and formed because of rapid molecular motion.

Less dense structure of ice leads that ice floats on water rather than sinks.

Hydrogen bonding

Considering CH_3OH, C_2H_6, Xe, and $(CH_3)_3N$, which can form hydrogen bonds with themselves? Draw the hydrogen-bonded structures.

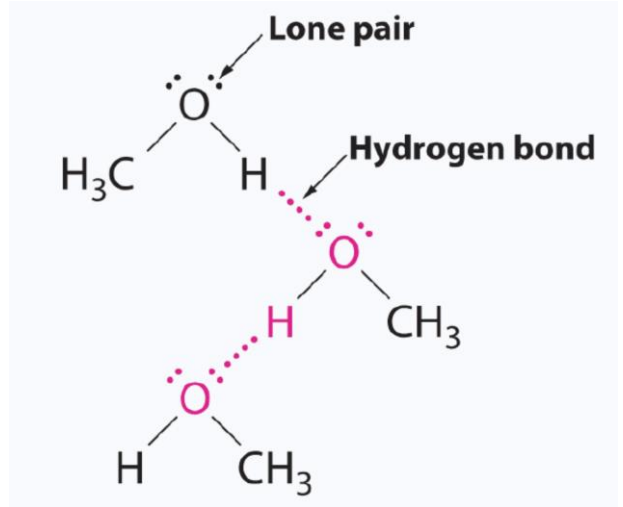
Hydrogen bonding

Considering CH_3OH, C_2H_6, Xe, and $(CH_3)_3N$, which can form hydrogen bonds with themselves? Draw the hydrogen-bonded structures.

Strategy:

1. Identify the compounds with a hydrogen atom attached to O, N, or F. (Possible hydrogen bond donors.)

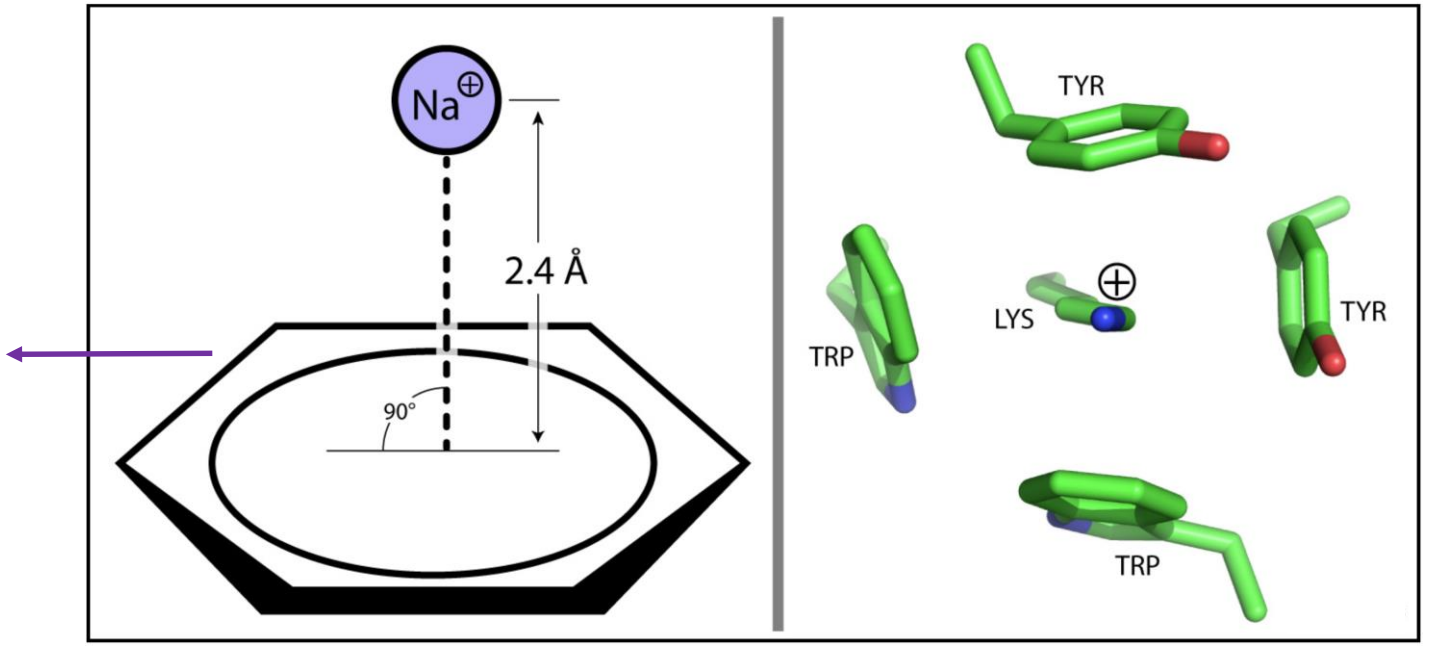
2. Of the compounds that can act as hydrogen bond donors, identify those that also contain lone pairs of electrons. (Possible hydrogen bond acceptors.)



Cation-π interaction

Strength ~ cation-dipole interactions between water and cations

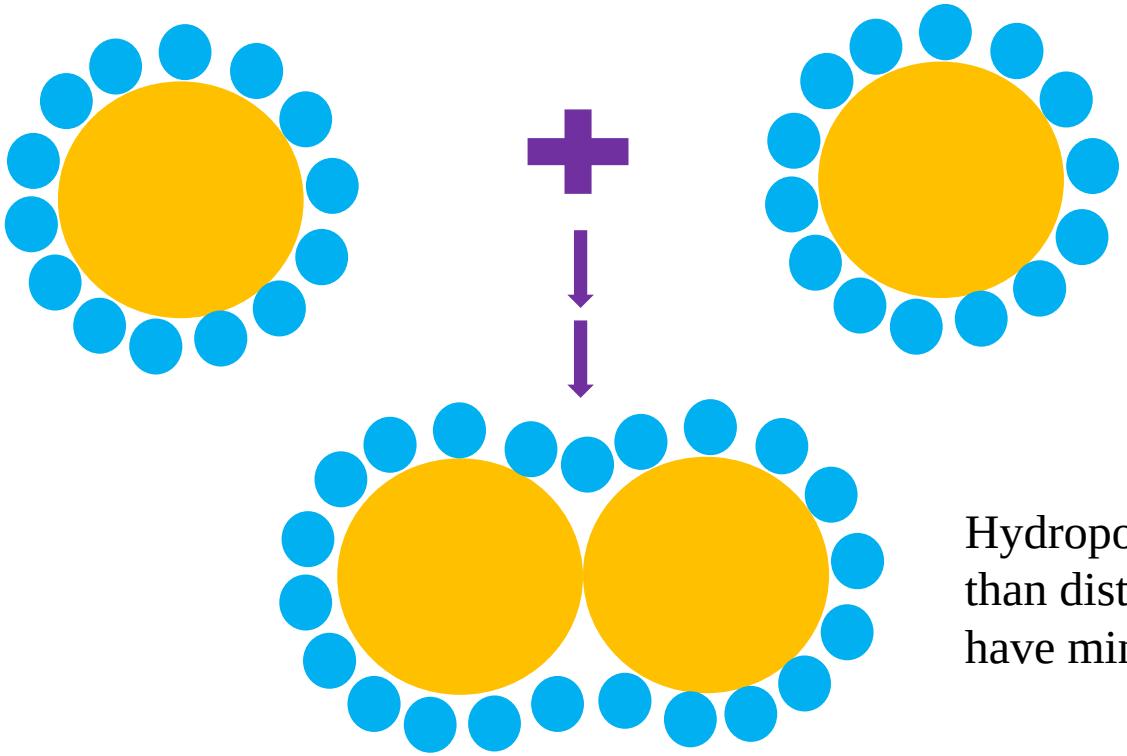
Partial negative charge above and below the π ring



The hydrophobic interaction

Describe the relations between water and **hydrophobes** (low water-soluble molecules)

Nonpolar molecules, usually have a long chain of carbons (R) that do not interact with water molecules





Hydropobes tend to clump up together rather than distributing itself in a water medium to have minimal contact with water

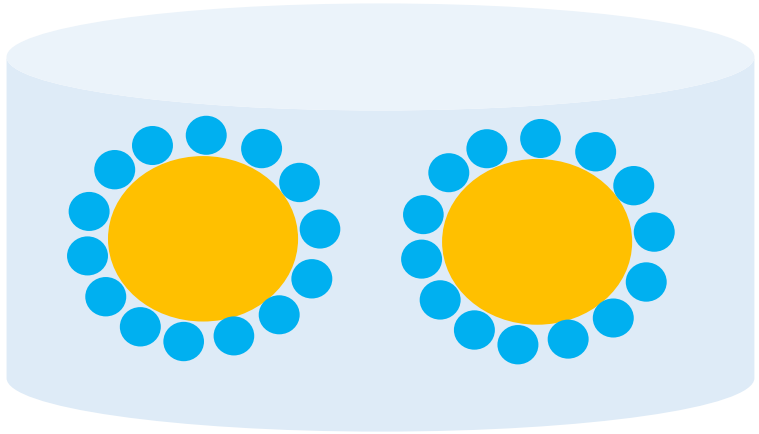
The hydrophobic interaction

Hydrophobic: With a positive $\Delta_{transfer}G$ from a nonpolar to a polar solvent

$$CH_4(\text{in } CCl_4) \rightarrow CH_4(aq)$$

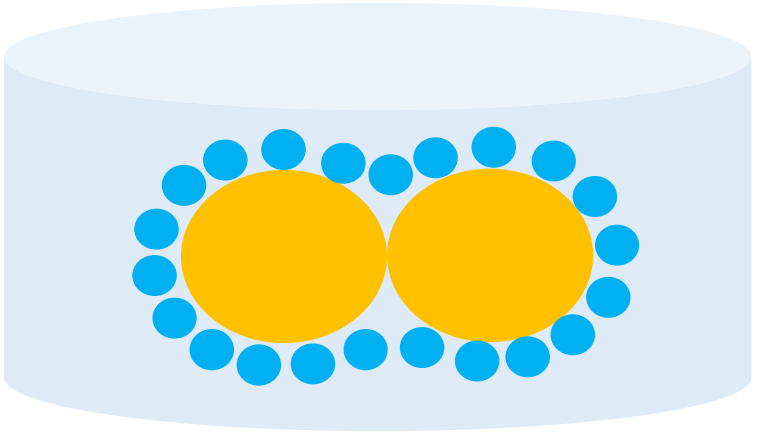
$\Delta_{transfer}\ G > 0$, $\Delta_{transfer}\ H < 0$, $\Delta_{transfer}\ S < 0$

No interaction



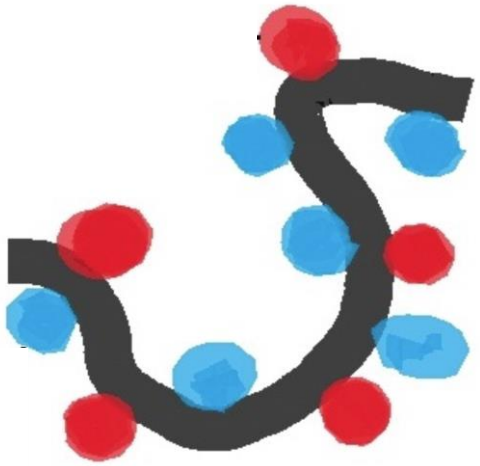
$\Delta_{transfer}\ H < 0$,
$\Delta_{transfer}\ S < 0$,
$\Delta_{transfer}\ G > 0$.

Hydrophobic interaction

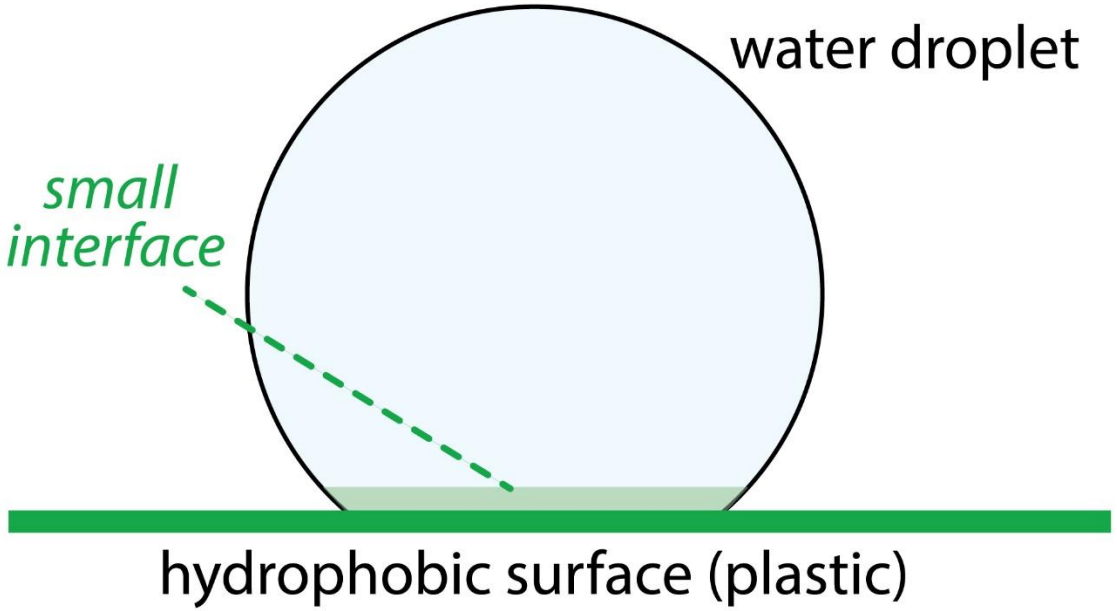


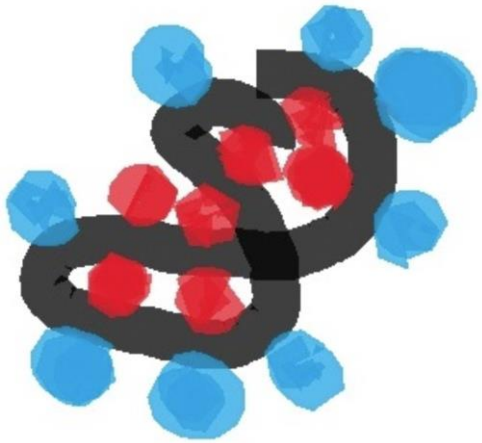
$\Delta_{transfer}\ H > 0$,
$\Delta_{transfer}\ S > 0$,
$\Delta_{transfer}\ G < 0$.

The hydrophobic interaction

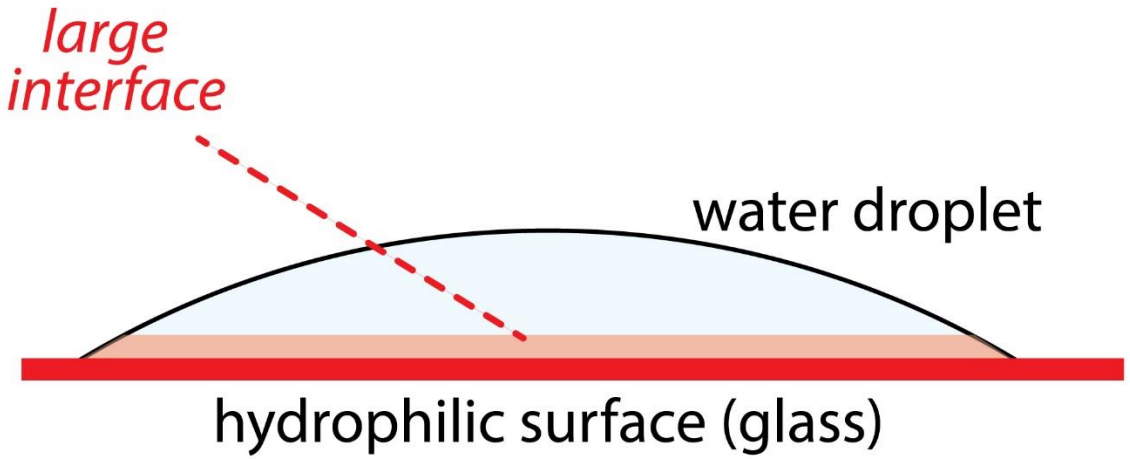


Isolated Protein





Protein in water



Strength of hydrophobic interactions

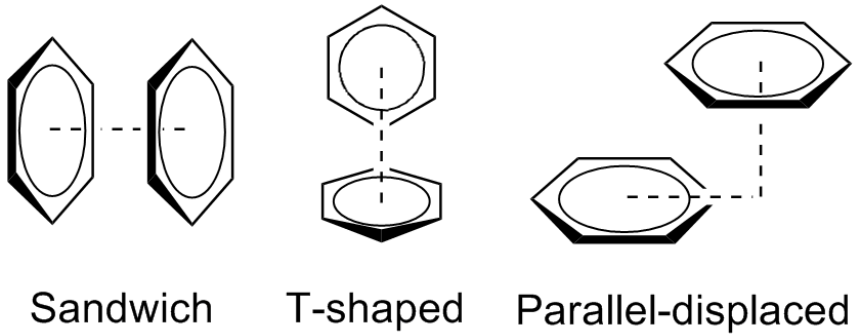
Stronger than van der Waals interactions or hydrogen bonds.

Depend on:

- Temperature: $T\uparrow \Rightarrow$ strength $\uparrow$ (not for extreme T);
- # of carbons: # $\uparrow \Rightarrow$ strength $\uparrow$;
- Shape: linear chain > branched chain;
- Aliphatic > aromatic;

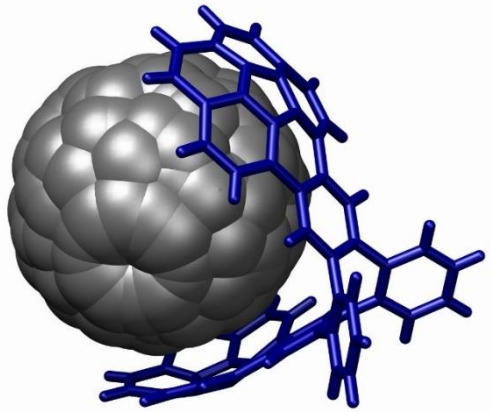
The π-π stacking

Attractive, noncovalent interactions between aromatic rings.

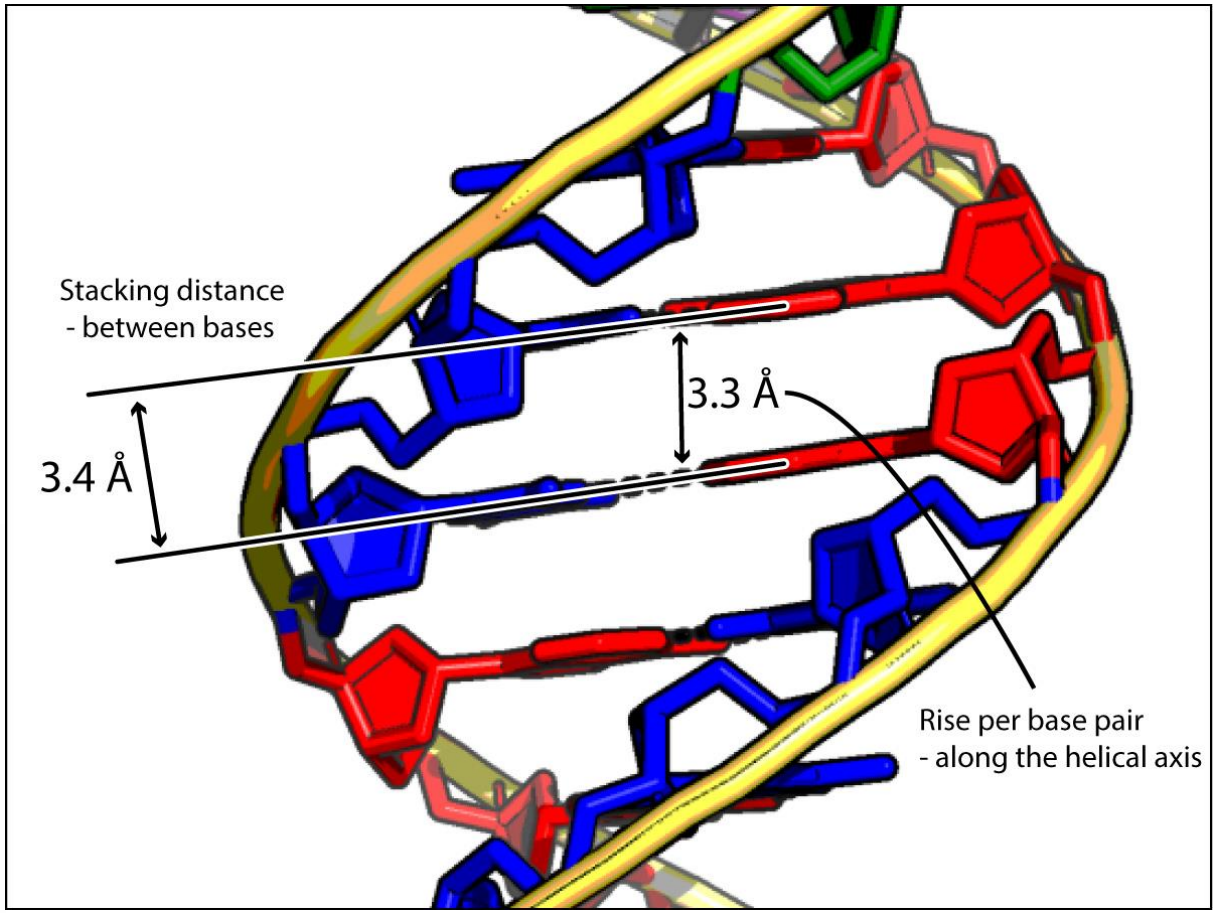


- Dispersion?
- Quadruple-quadruple interactions?
- Charge penetration?
- Hydrophobic effect?
- Aromaticity?

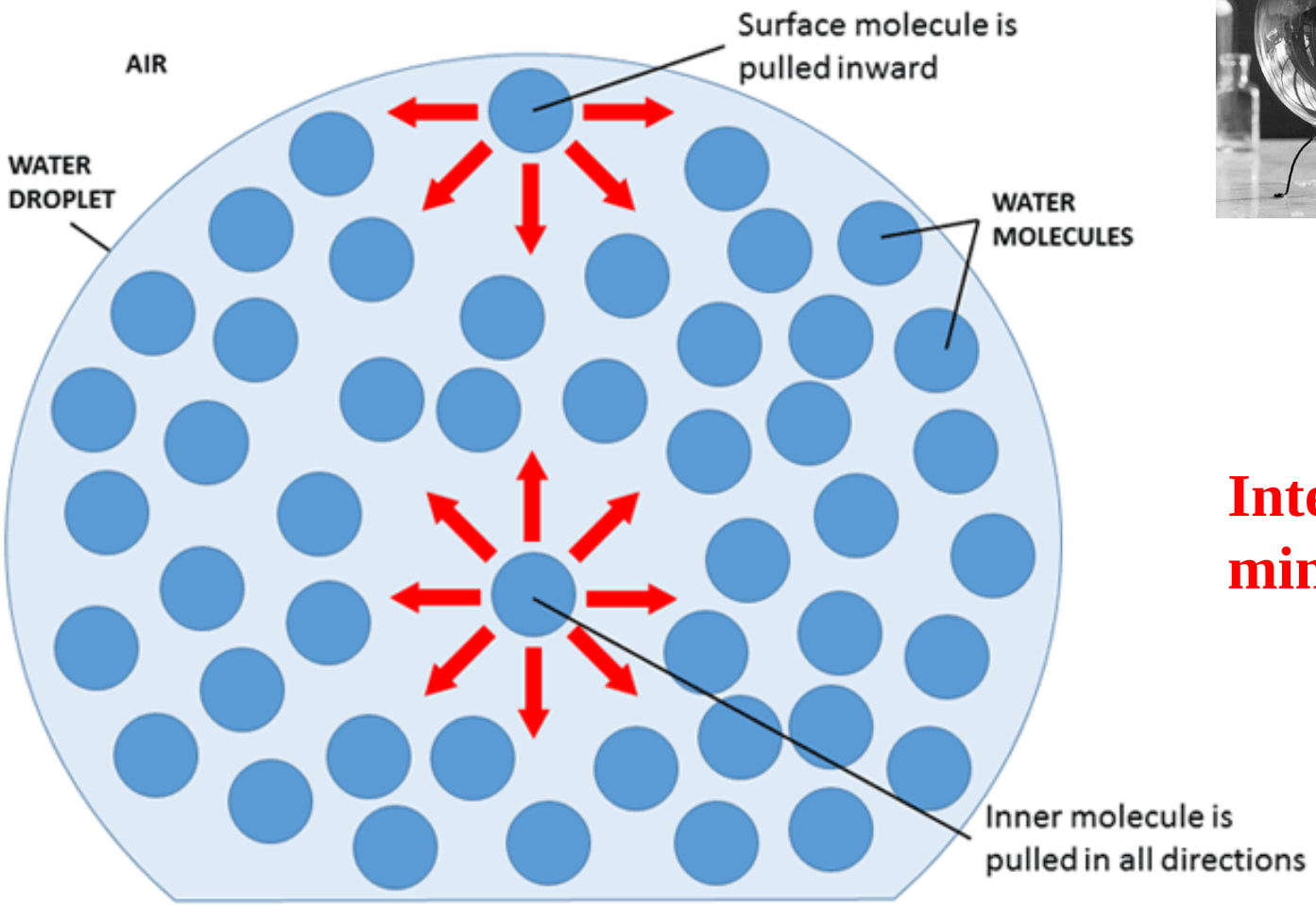
A fullerene bound in a buckycatcher through aromatic stacking interactions



J. Am. Chem. Soc. 2007,129,3842



Surface tension



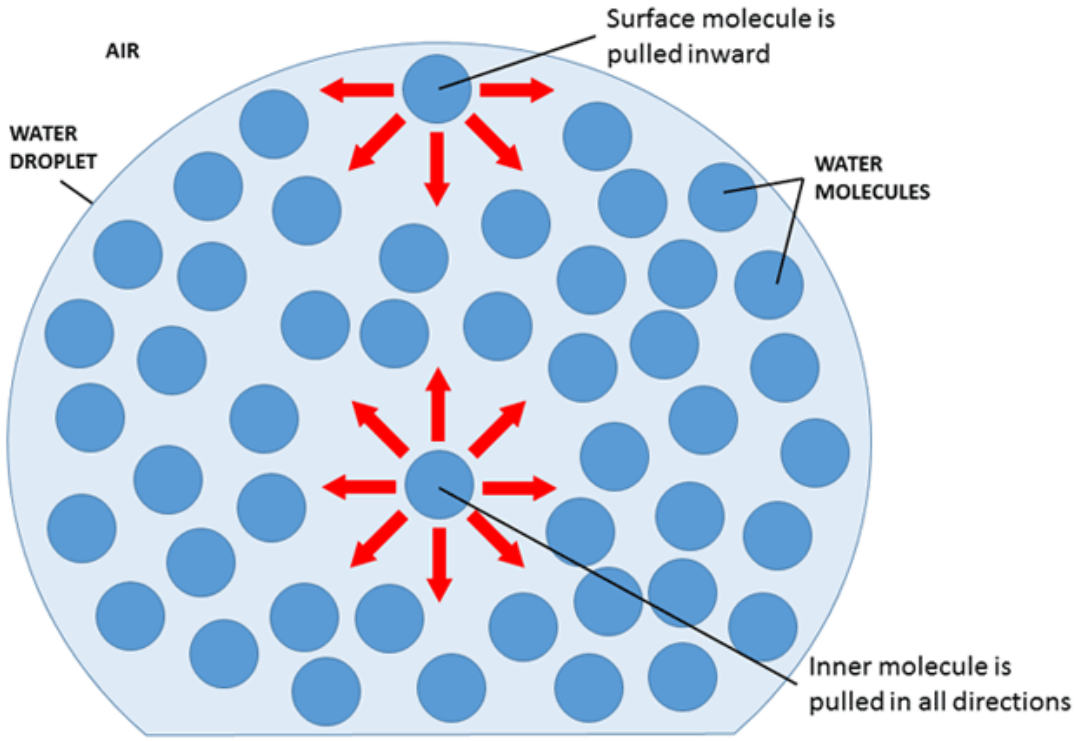






Intermolecular attractive forces act to minimize the surface area of a liquid!

Surface tension (表面张力)









Surface tension arises from the difference in the magnitudes of these forces within the liquid and at the surface.

Surface tension γ is the energy that must be supplied to increase the surface area by one unit.

Surface area

$$dw = \gamma d\sigma$$

Work

Free energy

$$dA = \gamma d\sigma \text{ at constant } V \text{ and } T$$

The SI units are J m^{-2} (or N m^{-1})

Surface tension



Mercury

substance	surface tension (dyne/cm)
water H(OH)	72.7
diethyl ether $(CH_3\text{-}CH_2)_2O$	17.0
benzene C_6H_6	40.0
glycerin $C_3H_2(OH)_3$	63
mercury (15°C)	487

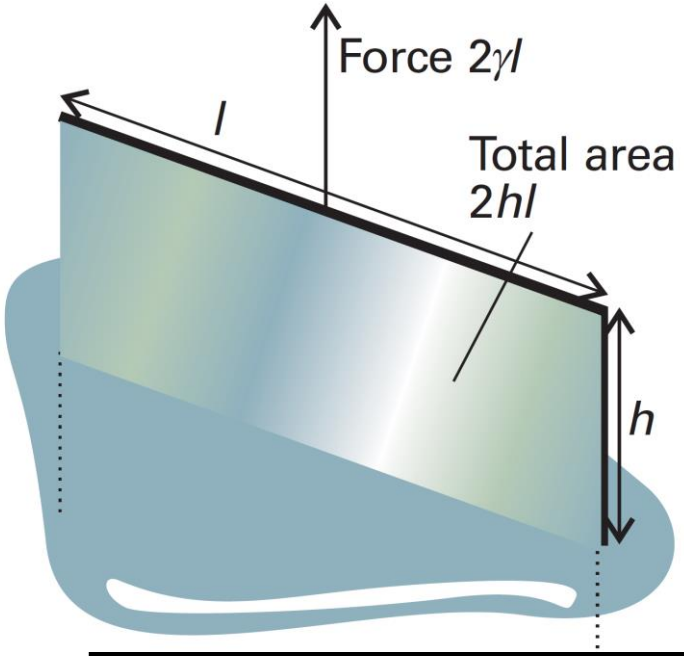
Mercury has the highest surface tension of any liquid at room temperature.

- Hydrocarbons and non-polar liquids have rather low γ;
- γ decreases with increasing T; (washing with warm water)
- γ is affected by the electrostatic charge of a body

"Mercury beating heart"

Surface tension

Calculate the work needed to raise a wire of length l and to stretch the surface of a liquid through a height. Disregard gravitational potential energy.



$$w = \gamma\sigma$$

$\sigma = 2hl$ (two surfaces, one on each side)

$$w = \underbrace{2\gamma l}_{\text{Force}} h$$

$\vec{\gamma}$ *is a force (per unit length) normal to the rod in the plane of the surface and directed toward the liquid.*

Curved surfaces



A **bubble** is a region in which vapour (and possibly air too) is trapped by a thin film; **two surfaces** (one on each side of the film.)

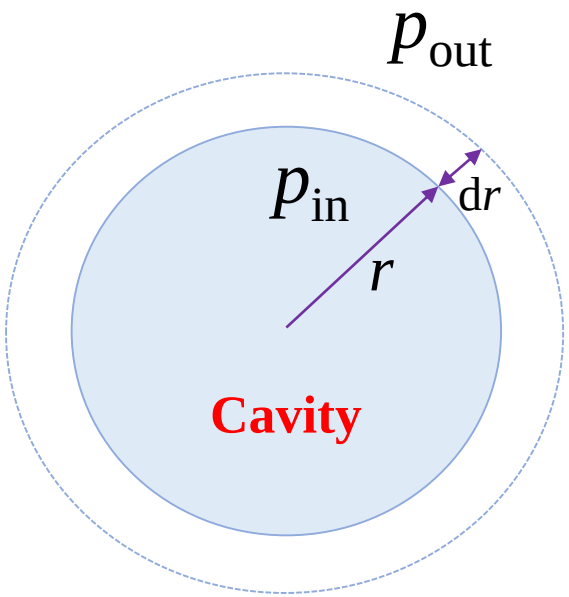


A **cavity** is a vapour-filled hole in a liquid; **one surface**.



A **droplet** is a small volume of liquid at equilibrium surrounded by its vapour (and possibly also air); **one surface**.

Laplace equation



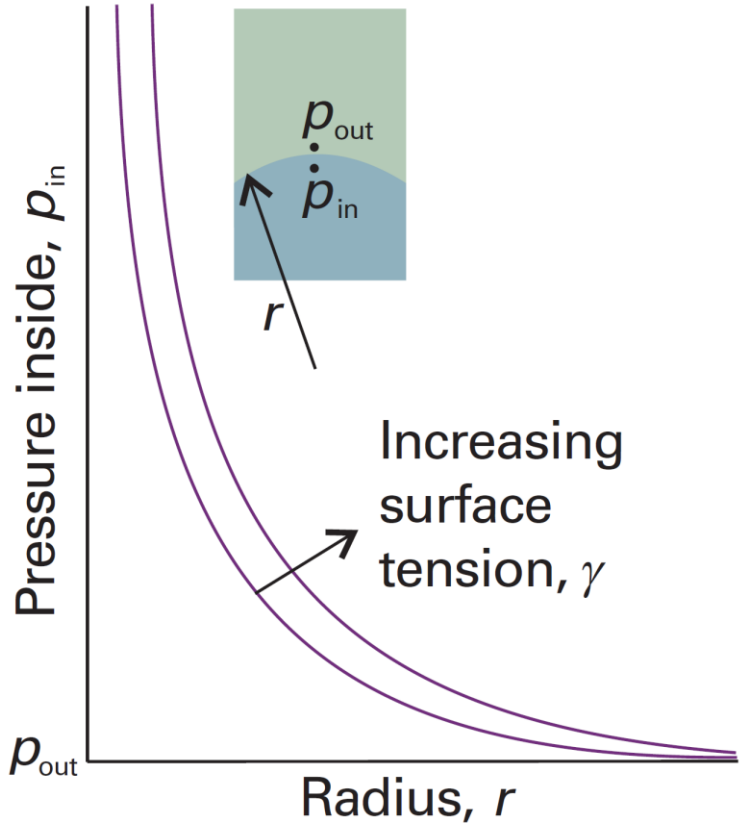
$$d\sigma = 4\pi(r+dr)^2 - 4\pi r^2 = 8\pi r dr$$

$$dw = 8\pi\gamma r dr = F dr$$

$$F = 8\pi\gamma r$$

At equilibrium, $4\pi r^2 p_{in} = 4\pi r^2 p_{out} + 8\pi\gamma r$

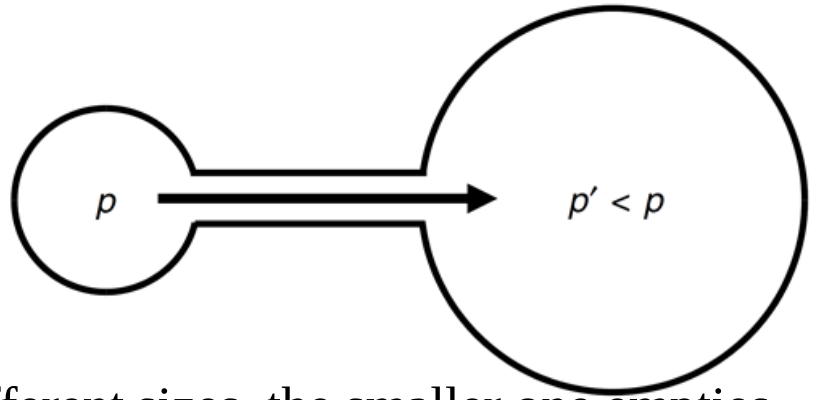
$$p_{in} = p_{out} + \frac{2\gamma}{r}$$



$r = \infty \Rightarrow$ flat surface

Laplace equation



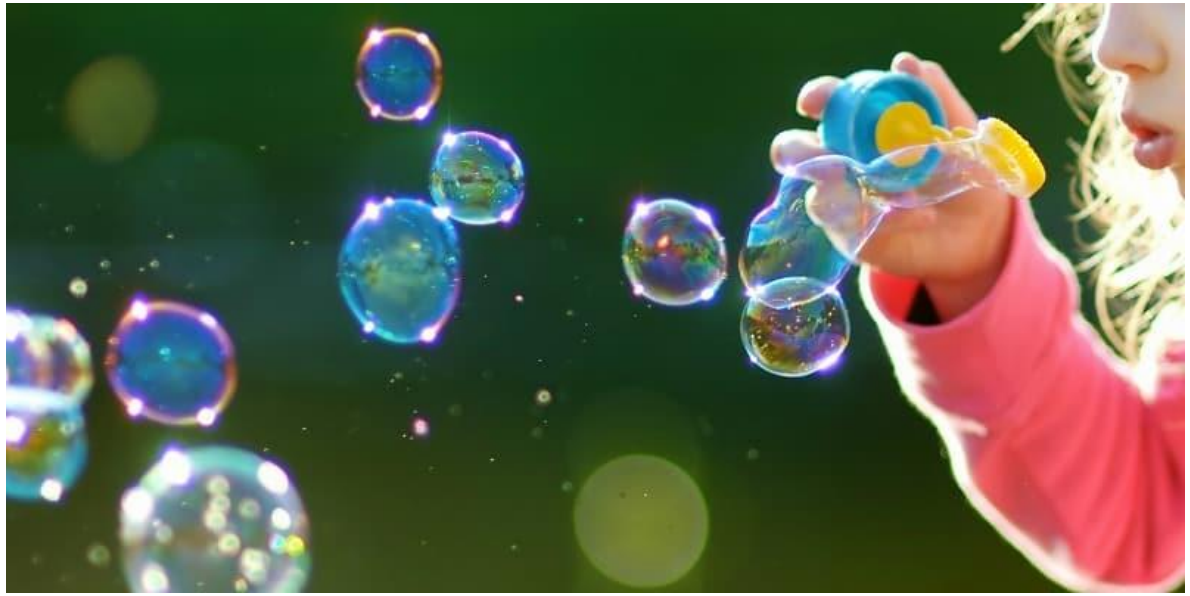


Connecting two bubbles of different sizes, the smaller one empties itself into the larger one.

— Ostwald's ripening

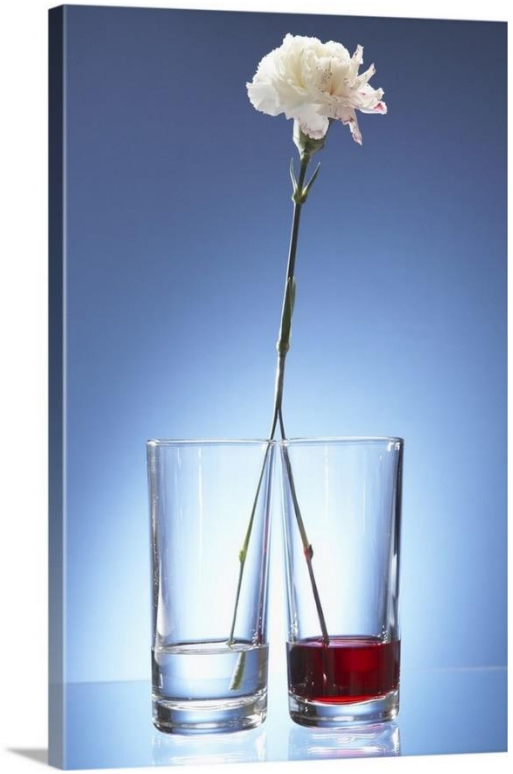
For a bubble,

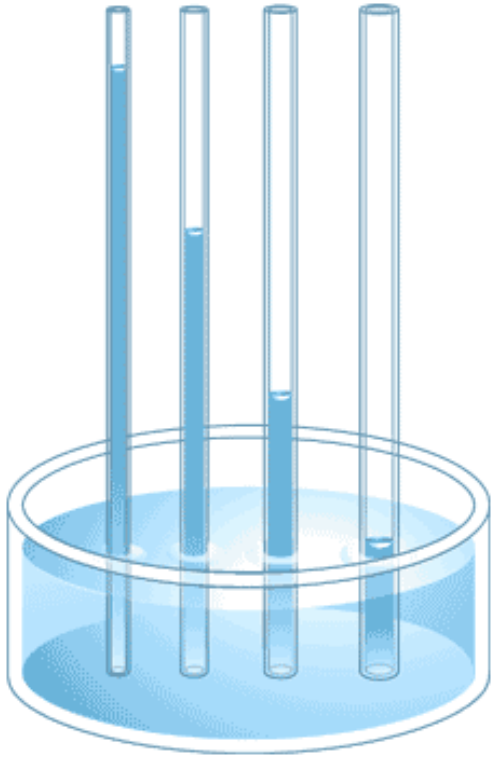
$$p_{\text{in}} = p_{\text{out}} + 4\gamma/r$$



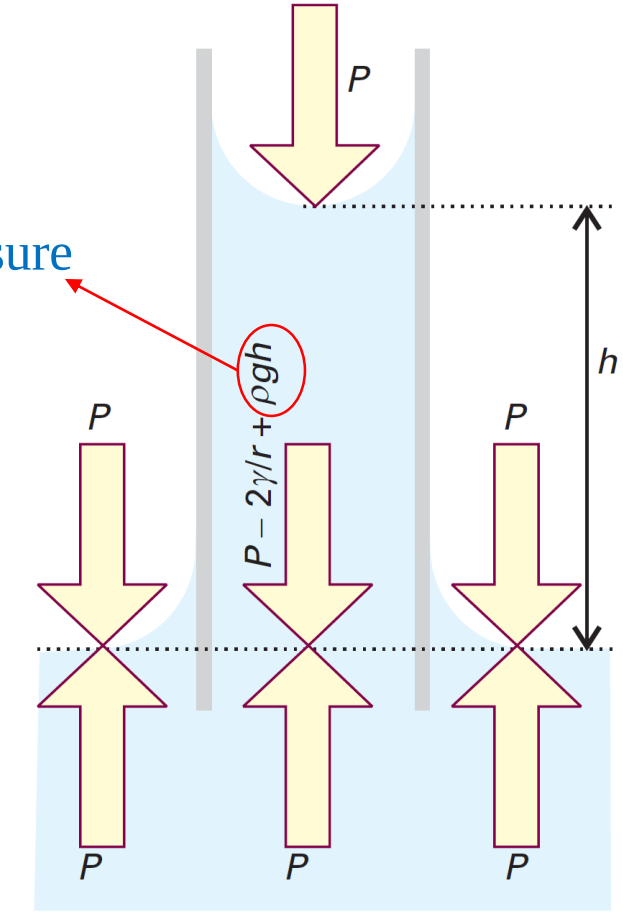
Capillary action (毛细作用)

- The tendency of liquids to rise up capillary tubes (tubes of narrow bore);
- A consequence of surface tension.
- The thinner the tube, the higher the liquid.

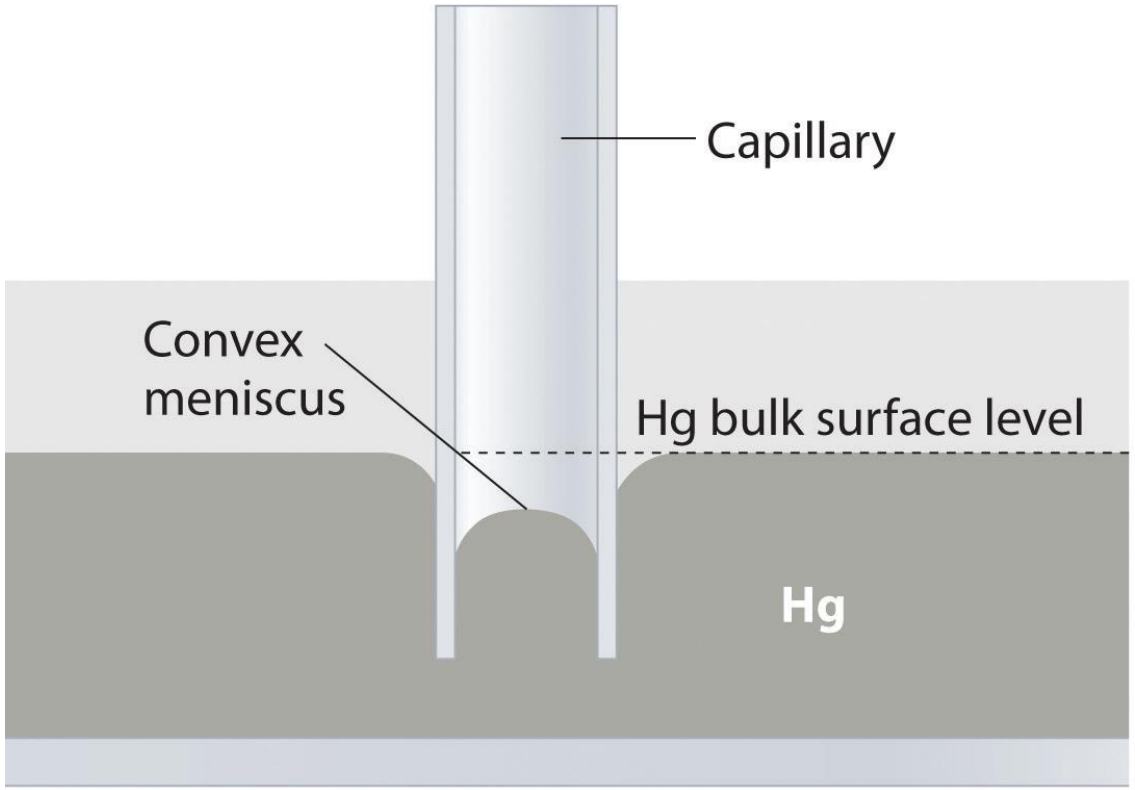


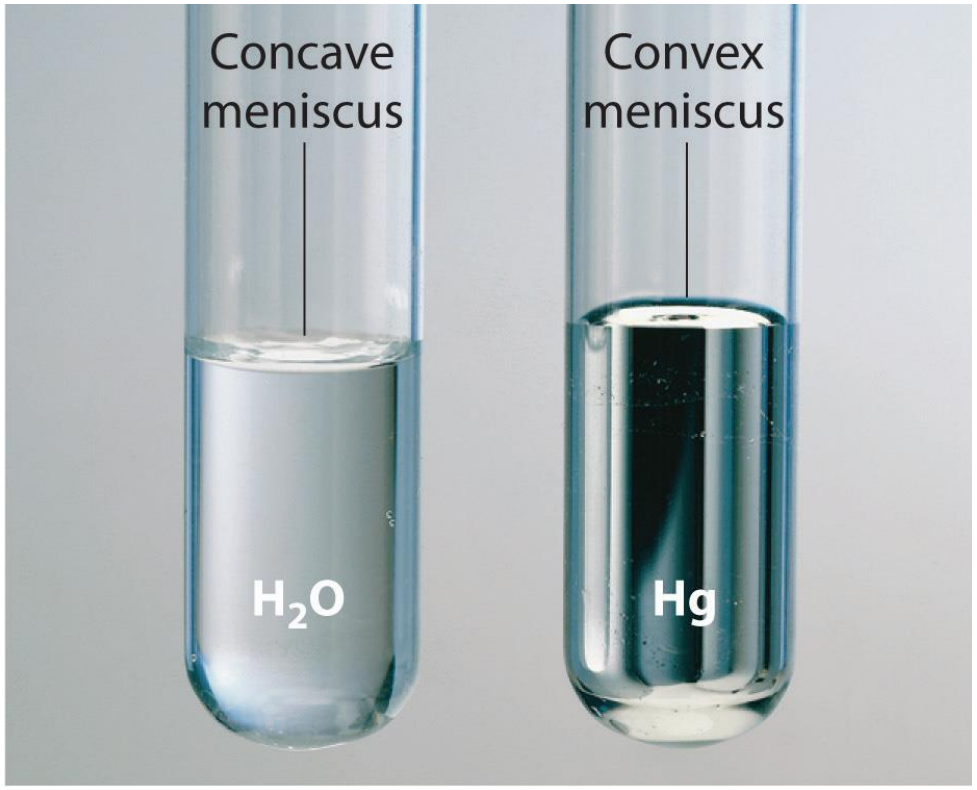


$$h = \frac{2\gamma}{\rho g r}$$



Capillary action



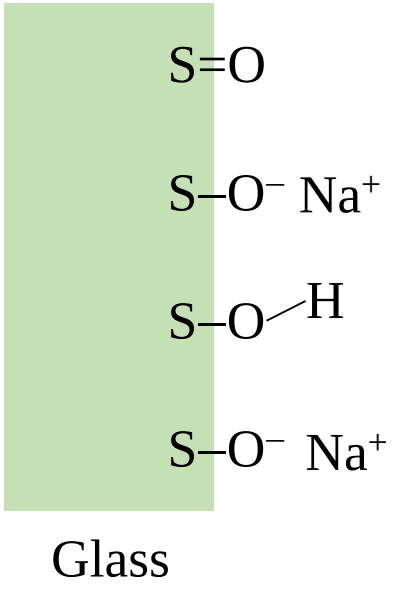


The adhesive forces between the liquid and the material of the capillary wall < the cohesive forces within the liquid ⇒ **Capillary depression**.

Tears of Wine

Induced by surface tension gradient





Both water and alcohol are hydrogen-bonding liquids.

$$\gamma_{water} > \gamma_{alcohol}$$

Volatility: alcohol > water

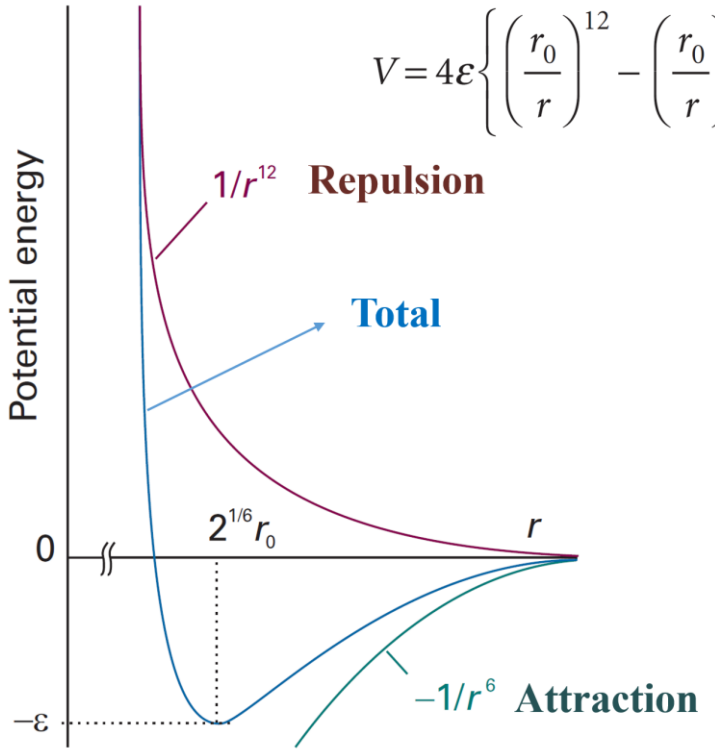
Review

A **van der Waals interaction** is the attractive interaction between closed-shell molecules.

- **Electrostatic force**: Dipole-dipole interactions
- **Induced force**: Dipole-induced-dipole interactions
- **Dispersion force**: Induced-dipole-induced-dipole interactions (**London interaction**)

Lennard-Jones potential

$$V = 4\varepsilon\left\{\left(\frac{r_0}{r}\right)^{12} - \left(\frac{r_0}{r}\right)^{6}\right\}$$



- Mass ↑, surface area ↑ → London force ↑;
- Depend on molecular shape

Hydrogen bonding A–H…B

The hydrophobic interaction

Depend on:

- Temperature: T↑ ⇒ strength ↑ (not for extreme T);
- # of carbons: # ↑⇒ strength ↑;
- Shape: Aliphatic > aromatic; linear chain > branched chain

Surface tension γ

Surface area

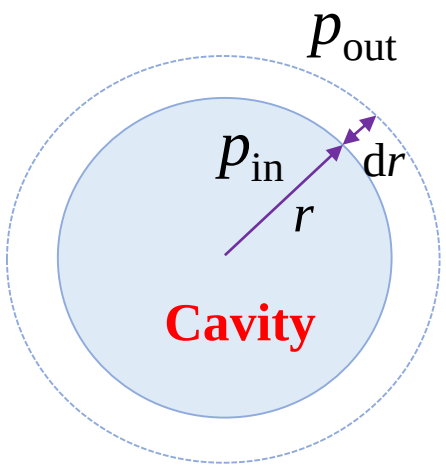
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Laplace equation

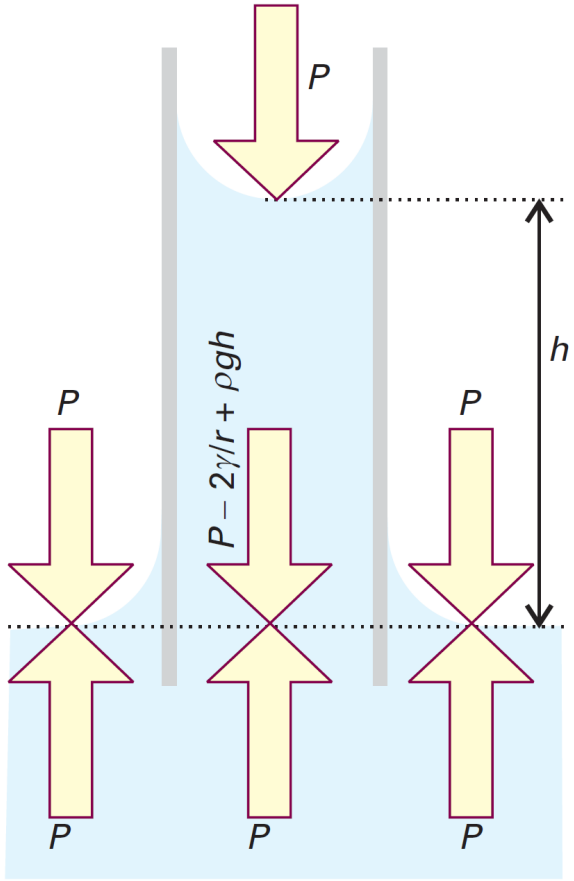


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For a bubble,

$$p_{in} = p_{out} + 4\gamma/r$$

Capillary action



$$h = \frac{2\gamma}{\rho g r}$$